

Troubleshooting Tools — Guided Notes

Unit 5.1.2 | CompTIA Network+ N10-009 Obj. 5.1 | N10-008 5.1

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

The `ping` command sends _____ Echo Request packets and listens for _____ replies. A 'Request timed out' means no reply came back, while 'Destination host unreachable' means _____ explicitly reported the host is not reachable.

Question 2

In ping output, the TTL value tells you approximately how many _____ the packet traversed. If you ping a Linux host and receive TTL=56, approximately _____ hops separated you from the destination.

Question 3

`tracert`/`tracert` maps the Layer 3 path by sending packets with incrementing _____ values. When a packet's TTL reaches zero, the router sends back a _____ message revealing its IP address.

Question 4

You run `nslookup portal.school.edu` and the Server line shows `10.1.1.5`. You then run `nslookup portal.school.edu 8.8.8.8` and get a different IP. What does this indicate, and what should you investigate?

Question 5

A workstation running Windows shows a 169.254.x.x address in `ipconfig /all`. This is called _____ and means the client _____ to receive a DHCP lease. List two possible causes.

Question 6

`netstat -an` shows the state LISTENING on port 443 and ESTABLISHED on port 3306. What do these states tell you, and what two services are most likely running?

Question 7

In Wireshark, _____ filters limit what is captured (set before capture begins), while _____ filters limit what is displayed from an existing capture. If you suspect ARP poisoning, what display filter and what pattern would you look for?

Question 8 — Real World Application

REAL WORLD: You are given a fiber optic cable from a patch panel to an AP that has no link light. You clean the connectors, verify the SFP types match, and verify the AP is powered. The link still won't come up. Name the hardware tool you use next, what it measures, and what result would tell you the cable has a break and where.

Download the answer key from the class server:

```
wget http://[SERVER]/resources/network-engineer/5-1-2-troubleshooting-tools/5-1-2-troubleshooting-tools-answer-key.pdf
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